

# Orexin population activity precisely reflects net body movement across behavioral and metabolic states

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## eLife Assessment

This **important** study shows that the activity of hypothalamic hypocretin/orexin neurons (HONs) correlates with body movement over multiple behaviors.

**Compelling** evidence, supported by sophisticated, cutting-edge tools and data analyses, highlights a link that appears to be unique to HONs. This work should be of interest to scientists studying peptidergic neurons, movement, energy regulation, and brain-body coordination.

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## Abstract

Tracking net body movement in real time may enable the brain to estimate ongoing demands and thus better orchestrate muscle tone, energy balance, and arousal. To identify neural populations specializing in tracking net body movement, here we compared self-initiated movement-related activity across genetically-defined subcortical neurons in the mouse brain, including dopaminergic, glutamatergic, noradrenergic, and key peptidergic neurons. We show that hypothalamic orexin/hypocretin-producing neurons (HONs) are exceptionally precise movement-trackers, encoding net body movement across multiple classified behaviors with a high degree of precision, independent of head acceleration. This tracking was so precise, that video analysis of the mouse body movement reliably served as a low-cost biometric for HON population activity. The movement tracking was independent of internal nutritional states, and occurred in a communication bandwidth distinct from HON encoding of blood glucose. At key projection targets, orexin/hypocretin peptide outputs correlated with self-initiated movement in a projection-specific manner, indicating functional heterogeneity in HON outputs. Finally, we found that body movement was not encoded to the same extent in other key neural populations related to arousal or energy. These findings indicate that subcortical orchestrators of arousal and metabolism are finely tuned to encode net body movement, constituting a bridge multiplexing ongoing motor activity with internal energy resources.

## Introduction

Neural activity related to body movement recently received renewed attention enabled by modern video tracking techniques<sup>1,2,3,4</sup>. Although the cortex is classically subdivided into sensory and motor regions<sup>5</sup>, movement-related signals make up a large proportion of single-neuron activity across both regions<sup>1-4, 6-18</sup>. Modern technologies for recording large numbers of neurochemically-undefined neurons also detected movement-related activity across subcortical brain areas<sup>1,2</sup>, but suggested some differences between brain areas<sup>3</sup>. This raises the question of whether population activities of neurochemically-distinct subcortical neurons are shaped by the magnitude and frequency of body movements.

We explored this question by examining the population activity of hypothalamic neurons producing the peptide transmitters hypocretins/orexins<sup>19, 20</sup>. Hypocretin/orexin neurons (HONs) track the body's glycemic dynamics and hunger states<sup>20, 21-23</sup>, and orchestrate arousal<sup>24-27</sup>. Metrics and interpretations of HON activity are of substantial interest in both basic and clinical sciences, due to their brain-wide influence and link to diagnosis and treatment of multiple brain disorders<sup>28-36</sup>. Yet, fundamental questions remain unresolved in relation to HONs and movement:

- Does HON population activity track specific behaviors, or the general magnitude of body movement?
- Given that hunger attenuates some neural operations<sup>37</sup>, does it disable HON movement tracking?
- Is movement vs glycemic information tracked by quantitatively distinct bandwidths of HON activity?
- How does HON tracking of body movement compare to other genetically-defined neural subcortical clusters linked to arousal and metabolism?

Here, we address these questions by simultaneously measuring population HON activity and body movement. We report that HONs encode the instantaneous magnitude of body movement; a feature that was observed across and within unique behaviors. HON movement encoding occurs in an activity bandwidth quantitatively distinct from their glucose-tracking bandwidth and is independent of metabolic or glycemic states. We further find that orexin/hypocretin peptide release dynamics related to body movement differ between various projection targets of orexin neurons. These results define a low-cost video biometric of HONs in a widely-used experimental setting, and add to our understanding of movement-related activation of genetically-distinct subcortical neural clusters.

## Results

### A general movement metric statistically explains rapid HON population dynamics across and within behaviors

To correlate HON dynamics with general body movement, we used a video camera to monitor  $n = 15$  head-fixed mice on a running wheel. Head fixation allowed us to repeatedly record movement from a fixed angle while focusing on factors other than head acceleration, as done in other recent studies<sup>1,2</sup>. Simultaneously, we recorded HON activity via intrahypothalamic fiber photometry of HON-selective GCaMP6s activity sensor (**Figure 1A**). Movement artifacts were controlled for via isosbestic excitation. To measure body movement, we derived a 1-dimensional movement metric by calculating the total per-pixel difference between consecutive frames, and then

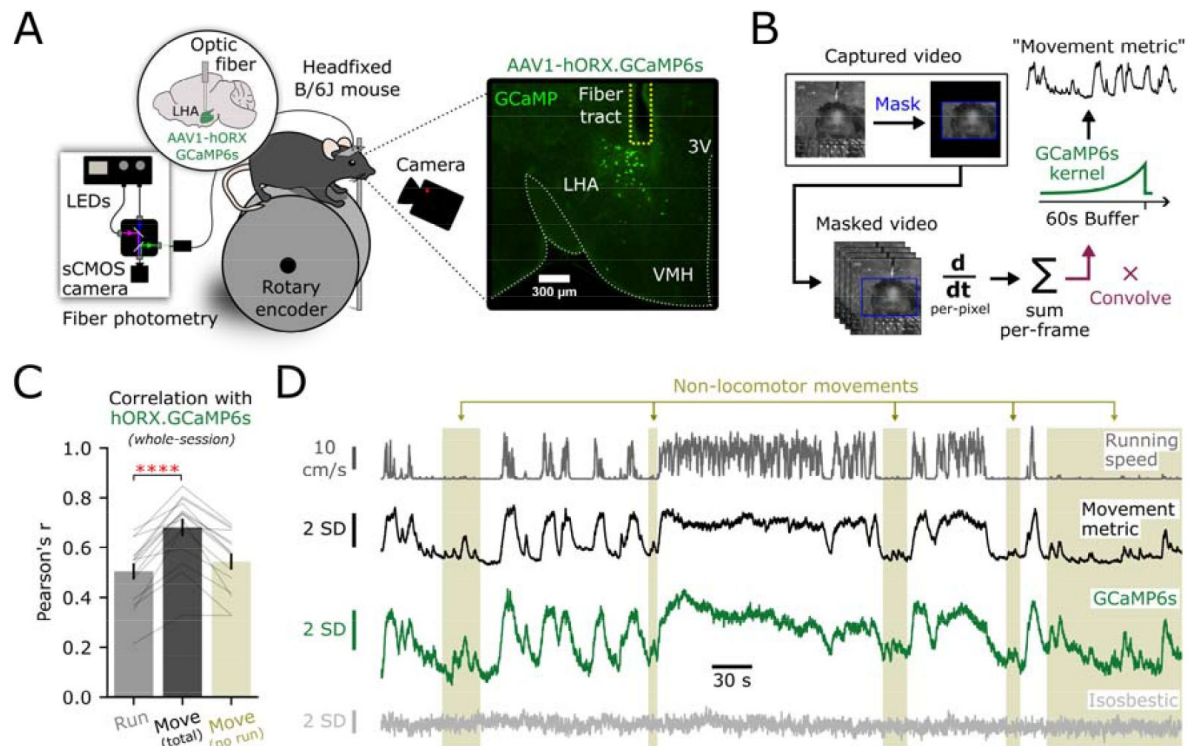
convolving with a decay kernel matched to the known dynamics of the GCaMP6s sensor (**Figure 1B**; <sup>38</sup>). We then quantified the correlation between recorded HON dynamics, movement, and running speed on the wheel. HON dynamics exhibited a positive correlation to both running-speed ( $r = 0.50 \pm 0.03$ ) and the movement metric ( $r = 0.68 \pm 0.04$ , **Figure 1C-D**). The movement metric's correlation was significantly stronger than running ( $p < 0.0001$ ). Notably, we also observed a positive correlation with the movement metric within epochs containing no locomotion on the wheel ( $r = 0.54 \pm 0.03$ ), demonstrating that movement encoding by HON dynamics is not limited to running (**Figure 1C,D**).

Does HON activity represent discrete behavioral states, or rather, encode movement magnitude generalized over behaviors? Both hypotheses could potentially explain the correlation in **Figure 1**. To compare our movement metric (**Figure 1B**) across different behaviors, we trained a deep-learning network (see methods) to classify video recordings into the five most commonly observed behaviors: resting, running, grooming, chewing, and sniffing (**Figure 2A**). Recorded HON activity was the highest during running, and lowest during resting showing significantly different activity levels across behaviors ( $p < 0.0001$ , **Figure 2B,C**). The average HON activity during specific behaviors followed a nearly 1:1 relationship against its average movement metric (**Figure 2D**). We then asked whether HON activity could be better explained by a classified behavior, or if it simply followed the movement metric regardless of the specific behavior. To test this, we fit three models in which HON activity was predicted by (1) the classified behavior, (2) the movement metric, or (3) the movement metric as an interaction effect with classified behavior (**Figure 2E**). Using the Akaike information criterion to assess model fit, we found model 1 was vastly inferior to model 3. However, the strength of this improvement was substantially less when comparing model 2 to model 3. These results suggest that HON activity is tracking the magnitude of movement change, not encoding different types of behavior. Finally, we sought to analyze HON dynamics during transitions between behavioral states. We first identified the six most common behavioral transitions (**Figure 2F**) and aligned them to both the movement metric and HON activity. As expected, the HON activity strongly resembled the magnitude of body movement exhibited along behavioral transitions (**Figure 2G,H**).

## Movement tracking within different frequencies and phases of HON population signal

Recorded HON dynamics varied over multiple timescales, exhibiting sub-second to several-minute oscillations. Due to this observation, we asked if HON encoding of movement was similar across frequency domains of HON activity. Empirical mode decomposition (EMD) adaptively decomposes a signal into a set of intrinsic mode functions (IMFs) which provide a time-frequency representation of the data, and is robust to the non-stationary and non-linear nature of biological signals<sup>39</sup>. Each IMF can be assigned a characteristic frequency which approximates the average oscillation in that signal component. Thus, we were able to “break down” hORX-GCaMP6s photometry into a set of simple IMFs, each capturing a different characteristic frequency of HON dynamics (**Figure 3A**). When plotting the average power of each HON IMF versus its characteristic frequency, we noted that maximum power generally occurred around 0.1 to 0.01 Hz (**Figure 3B**). A parallel EMD performed on the body movement metric revealed a maximum power in a similar frequency band (**Figure 3C**), suggesting that body movement and HONs have similar spectral profiles.

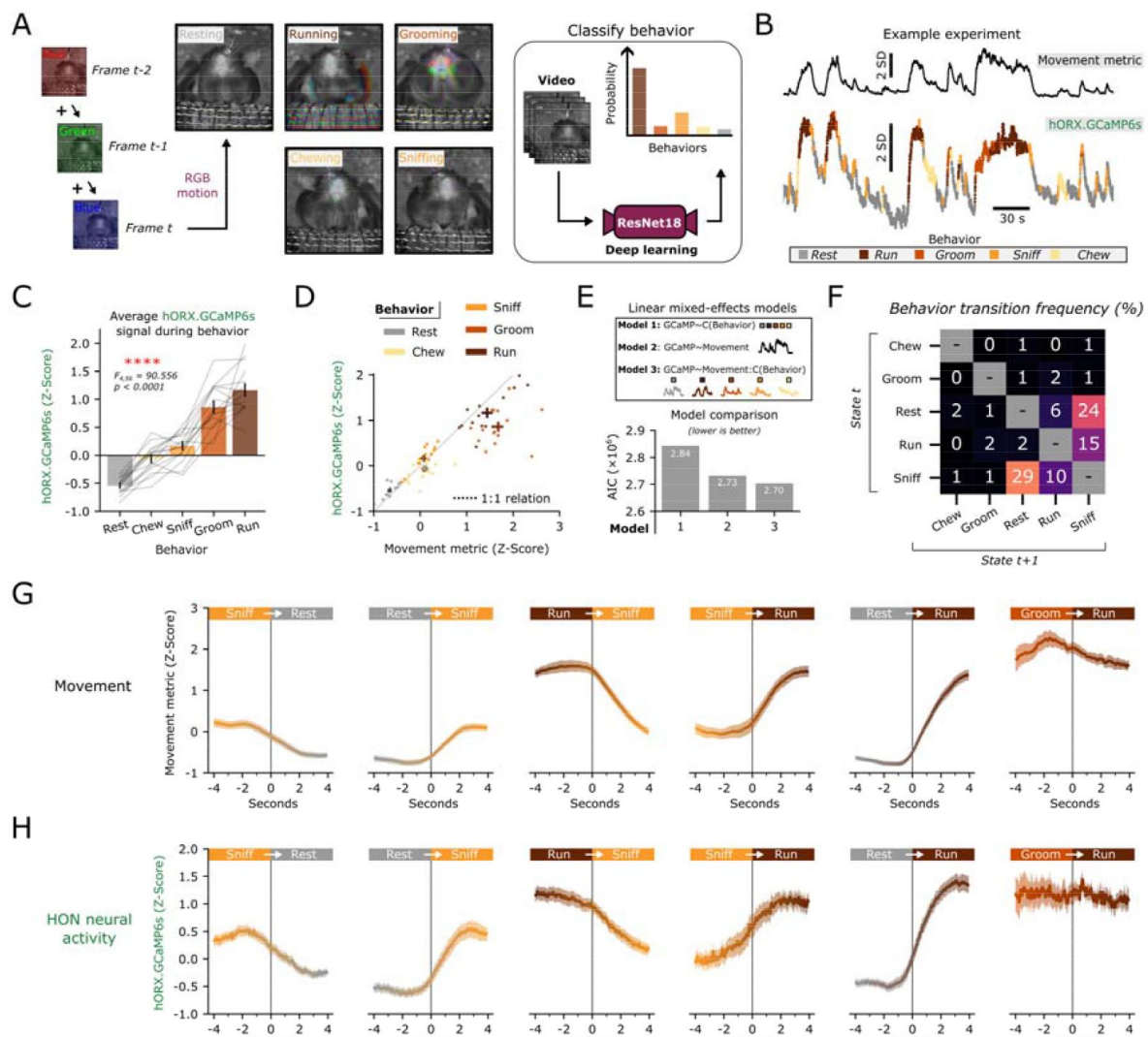
Do movements occur at specific phases within the HON activity rhythms? To examine this, we performed a phase preference analysis which tells us when, on average, a movement event occurs in the cycle of a HON oscillation. By convention, within each HON IMF, we defined the crest (“upstate”) to be at  $\pi/2$  radians and the trough (“downstate”) at  $3\pi/2$  radians. In turn, we split the movement metric (z-score) above and below zero to define “active” and “quiescent” movement epochs, respectively. We found that active epochs preferred the  $\pi/2$  HON phase while quiescent



**Figure 1.**

### Movement magnitude is represented by rapid HON dynamics

**A**, Schematic depicting the photometry system and voluntary wheel-running apparatus with video capture (left) and representative expression (right) of AAV1-hORX-GCaMP6s in HONs. LHA; lateral hypothalamic area; 3V; third ventricle; VMH; ventral medial hypothalamus. **B**, Processing pipeline to generate a "movement metric" from video recording of voluntary movement. **C**, Whole-session correlation of HON photometry with simultaneously recorded behavior metrics from  $n = 15$  mice: running ( $r = 0.50 \pm 0.03$ ), the movement metric ( $r = 0.68 \pm 0.04$ ), the movement metric in epochs where locomotion was  $< 1 \text{ cm} \cdot \text{s}^{-1}$  ( $r = 0.54 \pm 0.03$ ). Metrics given as mean  $\pm$  SEM, connected lines represent individual mice. Asterisks indicate comparison between running and movement metric (paired  $t$ -test:  $t_{14} = 6.676$ , \*\*\*\*  $p < 0.0001$ ). **D**, Traces from an example experiment. From top to bottom: running on the wheel, the video-derived movement-metric, HON GCaMP6s photometry, and isosbestic control. Shaded beige regions indicate non-locomotor epochs where the movement metric still reported strong positive correlations with HON activity.

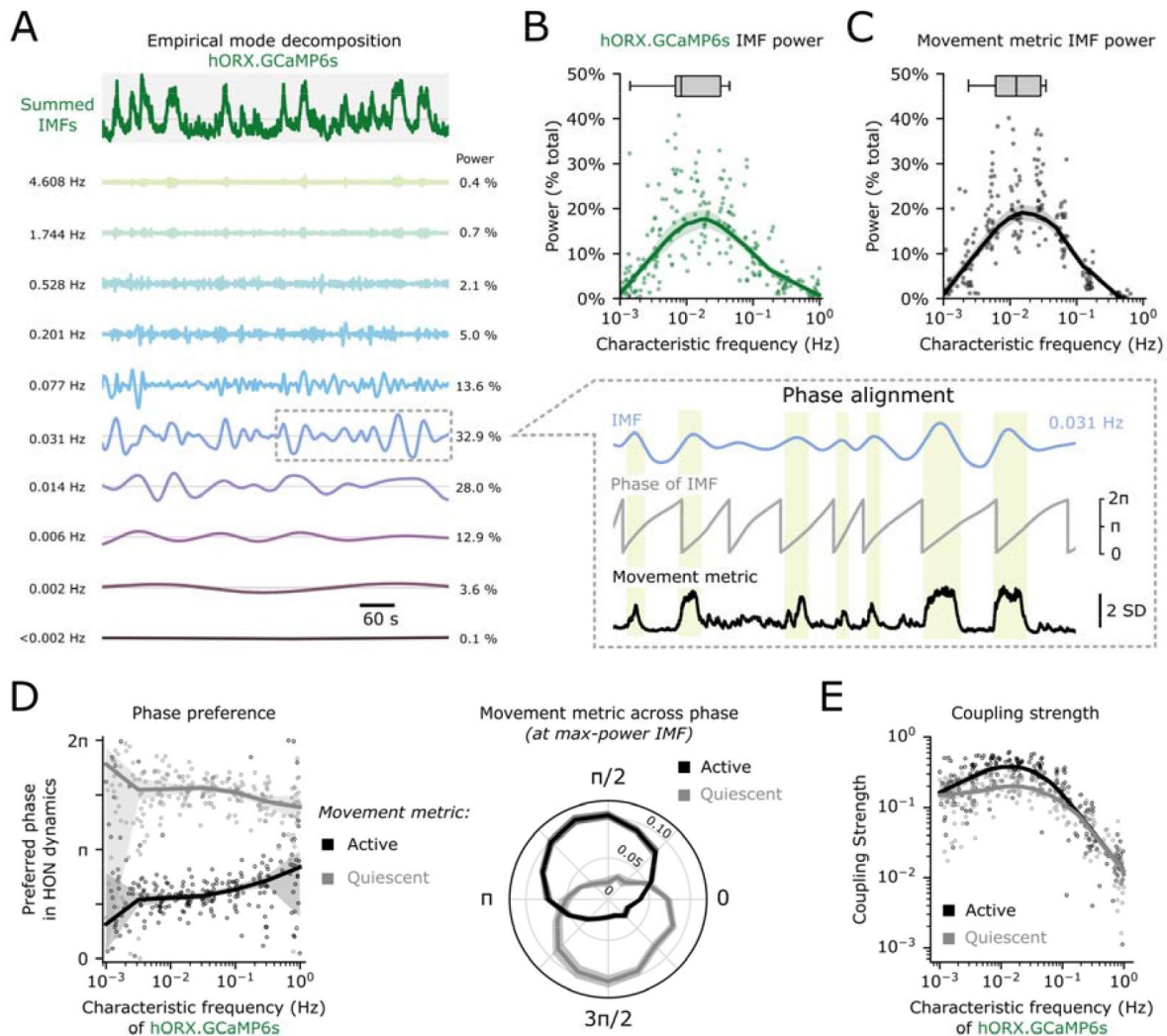


**Figure 2.**

### HON dynamics represent movement magnitude across classified behaviors

**A**, Diagram for classifying behaviors from video recordings via deep learning. **B**, Example experiments showing movement (upper, black) and color-coded photometry (lower) after behavioral classification. **C**, Average normalized HON GCaMP6s signal across classified behaviors in  $n = 15$  mice (rmANOVA:  $F_{4,56} = 90.556$ , \*\*\*\*  $p < 0.0001$ ). **D**, Average value of the normalized HON GCaMP6s signal from  $n = 15$  mice plotted against the average value of the movement metric in each classified behavior. Thick bars represent SEM across both metrics. The dotted line represents an idealized 1:1 linear relationship between movement and photometry with an intercept at zero. **E**, Comparison of three linear mixed effects models using movement and/or behavioral classification to predict HON GCaMP6s signal, quantified by Akaike information criterion. **F**, Average behavioral transition matrix for  $n = 15$  mice. **G**, Average movement metric aligned -4 to +4 seconds from a behavioral transition. The six most frequent behavioral transitions are plotted. Lines and shaded region represent mean and SEM. **H**, Same as G, but with hORX-GCaMP6s signal.





**Figure 3.**

### Movement magnitude is phase-aligned to HON activity across the frequency domain

**A**, Representative empirical mode decomposition (see methods) of HON GCaMP6s activity from one experiment. The top green trace displays a summation of all IMFs. Below are the first 10 IMFs sorted by characteristic frequency in Hz. Relative power is reported as a percentage of the total. Dashed grey box (middle right) represents a cutout of the 0.031 Hz IMF in which movement epochs (shaded beige regions) are clearly phase-aligned in the  $0-\pi$  period. **B**, Average power plotted against characteristic frequency of  $n = 229$  GCaMP6s-derived IMFs derived from 27 experiments using 15 mice. Thick line and shaded region represent a local regression and 95% CI. Boxplots represent the maximum power IMF from each experiment. **C**, Same as B, using  $n = 214$  movement metric-derived IMFs. **D**, Left: Preferred phase of both active (black) and quiescent (grey) epochs of the movement metric plotted against characteristic frequency. Dots represent  $n = 229$  GCaMP6s-derived IMFs derived from 27 experiments using 15 mice. Thick line and shaded region represent a circular regression and 95% CI. Right: average absolute value of the z-scored movement metric plotted on the radial axis against the phase of the maximum-power IMF. Lines represent mean  $\pm$  SEM from  $n = 15$  mice. **E**, Coupling strength of the movement metric to GCaMP6s IMFs in both active (black) and quiescent (grey) epochs plotted against the IMF's characteristic frequency. Dots represent  $n = 229$  GCaMP6s-derived IMFs derived from 27 experiments using 15 mice. Thick line and shaded region represent a local regression and 95% CI.

epochs preferred the  $3\pi/2$  HON phase, especially in the 0.1 to 0.01 Hz frequency domain (**Figure 3D**). Strongest phase coupling between HONs and movement was also observed in this frequency domain (**Figure 3E**).

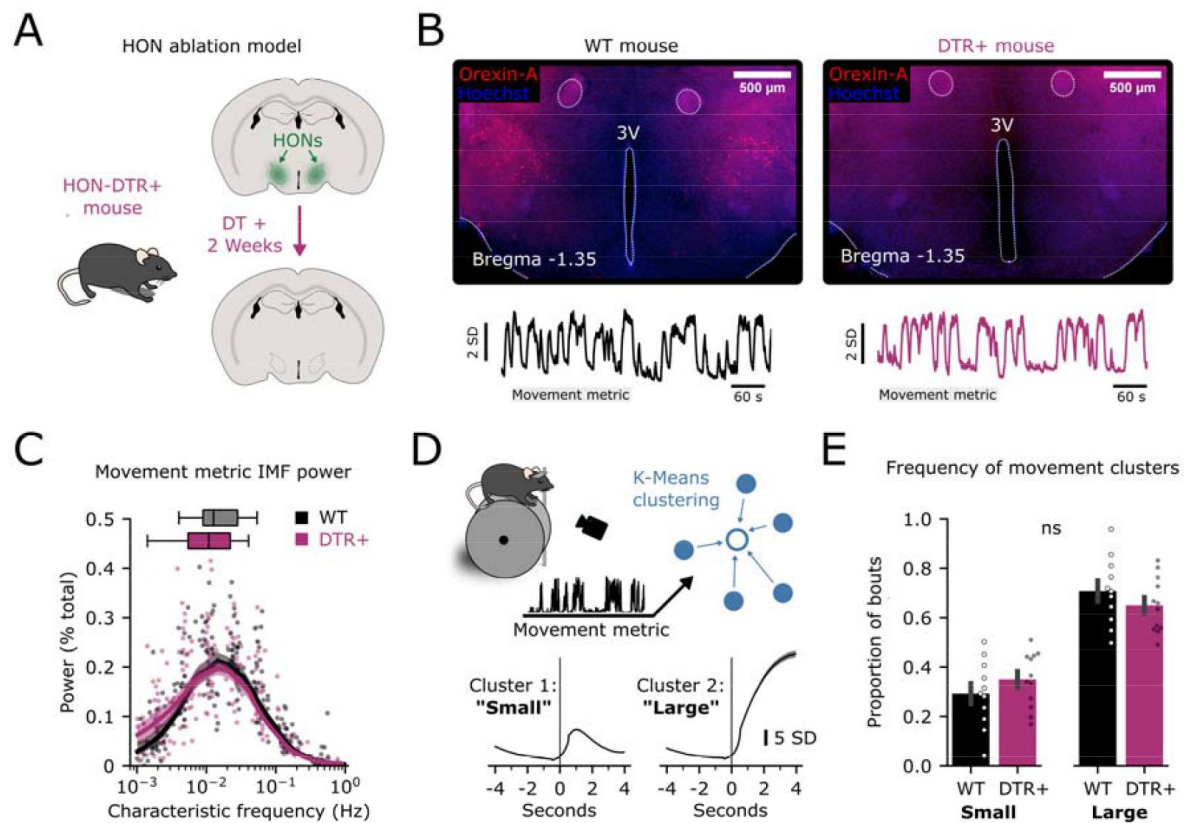
Finally, we sought to confirm whether HON simply tracked, or also shaped, the body movements characterized in **Figure 3C**. We reasoned that if HONs only tracked the movements, then HON ablation would not substantially alter the power spectrum of the movements; conversely, if HONs shaped the movements, then HON ablation would alter the movements. We selectively ablated HONs using an extensively validated HON-DTR mouse model<sup>23, 40, 41</sup> (**Figure 4A,B**). Maximum-power IMFs were found in a similar frequency range peaking around 0.1-0.01 Hz in HON-ablated and control mice (**Figure 4C**), suggesting HONs are not necessary to maintain movement frequency profiles. To further explore this using an alternative analysis, we used a K-Means clustering algorithm to sort self-initiated movements into two broad categorical clusters constituting “small” and “large” movements (**Figure 4D**). The frequency of “small” and “large” movements was not different across control and HON-ablated mice (**Figure 4E**). Overall, these results define frequencies and phases of HON activity that track body movement information.

## Orexinergic representation of movement at different HON projection targets

Monoaminergic nuclei producing dopamine and noradrenaline, substantia nigra pars compacta (SNc) and locus coeruleus (LC) respectively, have long been studied for their involvement in arousal and movement<sup>42, 43</sup>. Both are densely innervated by HON axons and express orexin receptors<sup>29, 44</sup>. Do HONs transmit movement information equally to these projection targets? To investigate this, we expressed the genetically-encoded orexin peptide biosensor OxLight1<sup>45</sup> into the SNc and LC for simultaneous dual-site recording in the same mouse (**Figure 5A-C**). We found that the “active” epochs of the movement metric differed in phase preference to OxLight1 signal between the brain regions: SNc orexin recordings showed a phase preference to the “downstate” ( $3\pi/2$ ) phase, while LC orexin recordings preferred the “upstate” ( $\pi/2$ ) phase, particularly in the 0.1 to 0.01 Hz frequency band (**Figure 5D**), with the highest phase coupling in the lower frequency bands (**Figure 5E**). This suggests that orexin release may heterogeneously represent body movement at different HON projection targets. When orexin dynamics were aligned to “small” and “large” movement clusters, we observed rising signals in the LC, but falling signals in the SNc, which was especially apparent in association with “large” clustered movements (**Figure 5F**). Together, these analyses indicate that orexin peptide release related to movement is non-homogenous across two major HON projection targets.

## HON movement tracking across metabolic states

Given the robust correlation of HON activity and movement across the behavioral and frequency spectra, we sought to contrast our findings against another major physiological factor thought to influence HON activity, the metabolic state<sup>20–22</sup>. First, we tested whether movement encoding by HONs depended on fasting state, which is known to strongly modulate HONs<sup>20, 46</sup>. We found that correlations of HON activity and movement were not significantly different in *ad-libitum* fed, 18h-fasted then re-fed, and 18h-fasted mice (**Figure 6A**). Second, we sought to quantify how the HON movement encoding compares to the recently-described HON encoding of lower-frequency blood glucose information, namely the negative relations between HONs and first temporal derivative of blood glucose<sup>23</sup>. Using a small number of mice from this pre-existing dataset<sup>23</sup> in which video recordings were also available (**Figure 6B**), we again performed an EMD to align individual HON IMFs to both movement and the first derivative of blood glucose after intragastric infusion of glucose (0.24 g/mL, 900  $\mu$ L over 10 minutes; **Figure 6C**). To explore and formally define the HON signal frequency bands that carry information about movement and blood glucose, we fit each HON IMF using a multivariate model with glucose derivative and

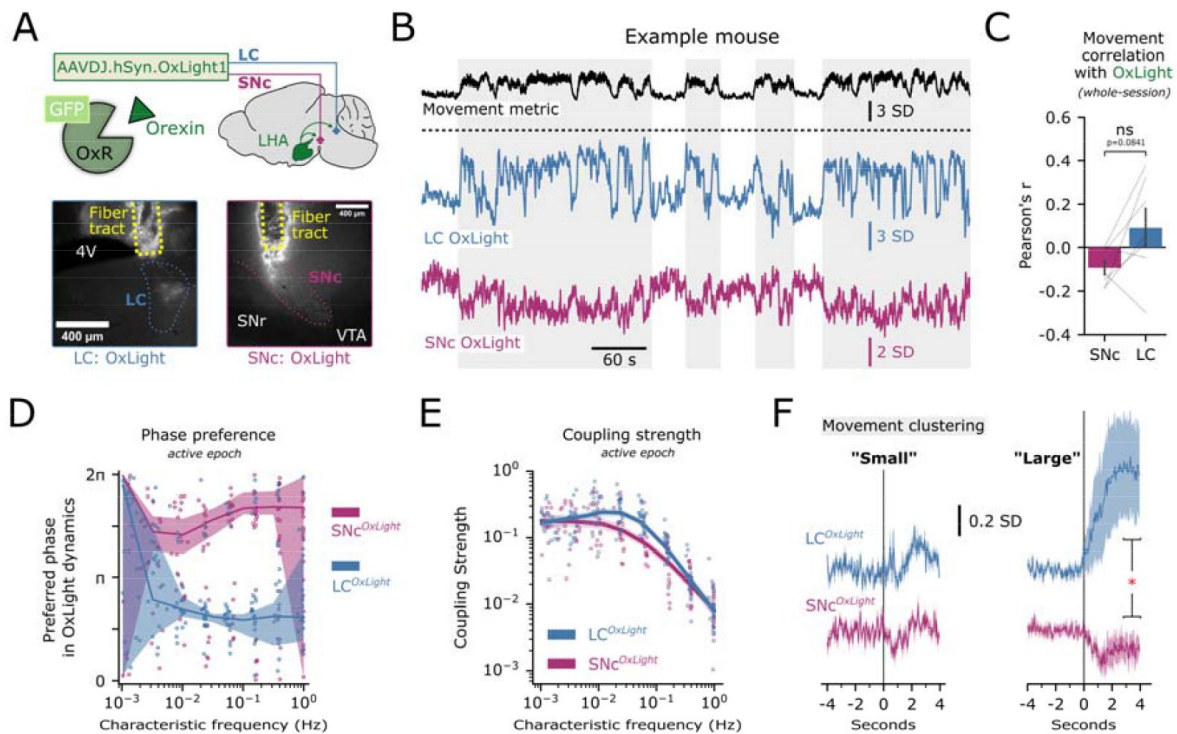


**Figure 4.**

### HONs are not required to maintain movement profiles

**A** HON ablation in HON-DTR+ mice. **B**, Brain slice histology of a wild-type mouse (WT, left, image representative of  $n = 11$  mice) and a HON-ablated mouse (DTR+, right, image representative of  $n = 12$  mice) stained for OrexinA (red). 3V = 3rd ventricle. **C**, Average power plotted against characteristic frequency of movement metric-derived IMFs (WT:  $n = 217$  IMFs from 30 experiments using 11 mice. DTR+  $n = 267$  IMFs from 34 experiments using 12 mice). **D**, Diagram of K-Means clustering to generate two clusters of movement representing voluntary initiation motion following  $> 4$  s of no motion, see methods. **E**, Proportion of bouts which were assigned "small" or "large" were not significantly different across genotypes (Unpaired  $t$ -test,  $t = 1.1167$ ,  $p = 0.2767$ ).





**Figure 5.**

### Movement-related orexin peptide release is non-homogenous across projection targets

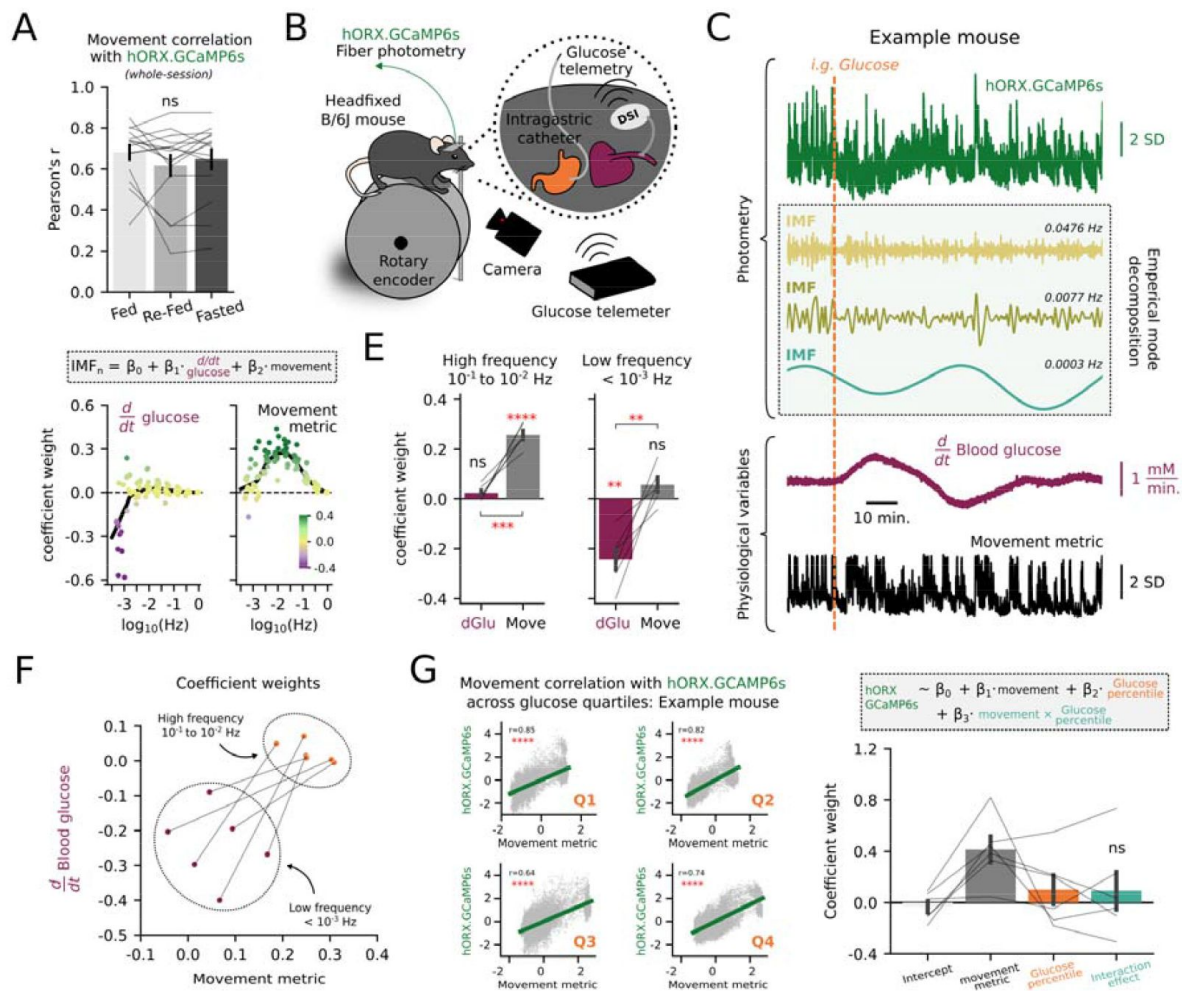
**A**, Schematic representing targeting of the orexin sensor OxLight1 to the LC and SNc, representative histology from  $n = 7$  mice. **B**, Example experiment displaying the movement metric and OxLight1 signal in the LC (blue) and SNc (purple) in the same mouse. Shaded regions represent movement epochs. **C**, Mouse-averaged whole-session correlation of movement with OxLight1 photometry in the SNc and LC (paired  $t$ -test:  $t_6 = 2.068$ ,  $p = 0.0841$ ). **D**, Preferred phase of individual EMD-derived OxLight1 IMFs to active epochs of the movement metric plotted as a function of the IMF's characteristic frequency.  $n = 317$  IMFs from 21 experiments using 7 mice. Thick line and shaded region represent a circular regression and 95% CI. **E**, Coupling strength of individual EMD-derived OxLight1 IMFs to active epochs of the movement metric plotted as a function of the IMF's characteristic frequency.  $n = 317$  IMFs from 21 experiments using 7 mice. Thick line and shaded region represent a local regression and 95% CI. **F**, Photometry baselined -3 to -1 seconds before movement initiation in two clusters: small movements (left) and large movements (right). Lines and shaded regions represent mean  $\pm$  SEM of  $n = 6$  mice. Asterisk represents a paired statistical comparison of mean signal 3 to 4 seconds after movement initiation (Wilcoxon test,  $D_6 = 1.0$ , \*  $p = 0.03125$ ).

movement as weighted predictors of each IMF. We plotted the two fitted weights against corresponding IMF frequency (**Figure 6D**). Large negative weights for the blood glucose derivative were assigned to lower frequency IMFs, while large positive weights for movement were assigned to higher frequency IMFs (**Figure 6D**). When binning across frequencies, statistical comparison of these weights indicated that movement had significantly different weights than glucose derivative both in the 0.1 to 0.01 Hz and <0.001 Hz frequency bands (**Figure 6E**). Notably, glucose derivative did not have significant weights in the higher frequency band, whereas movement did; conversely, movement did not have significant weights in the lower frequency band, whereas glucose derivative did (**Figure 6E**). Overall, these data indicate that movement representation in the HON population signal is spectrally orthogonal to fasting state or blood glucose dynamics, and is carried selectively in the high-frequency (0.1-0.01 Hz) band of the signal, while glucose information is contained in lower frequency (<0.001 Hz) band (**Figure 6F**). Finally, we examined if movement encoding by HONs changed as a function of the blood-glucose percentile. By examining the interaction term from linear regressions fit to the  $n = 6$  mice, we observed that movement encoding by HONs was broadly independent of blood glucose percentile (**Figure 6G**).

Outside of metabolism, it has long been known that HON activity co-varies with arousal<sup>47, 48</sup>. We therefore wondered how HON movement encoding compares to their encoding of arousal. To explore this, we quantified arousal by tracking pupil size (an established metric of arousal in both mice and humans<sup>49, 50</sup>). In these recordings we also quantified ocular movements. We then performed multivariate analysis to disentangle and compare statistical contributions of arousal vs body and eye movements to HON activity (**Figure 7A**). In confirmation of previous results<sup>51</sup>, simple regression confirmed that pupil size was positively correlated with HON activity (**Figure 7B**,  $r = 0.45 \pm 0.08$ ). Ocular movements had a smaller, but still positive correlation ( $r = 0.26 \pm 0.02$ ). Using drop-one feature analysis on a partial least squares regression, we found movement was the most important (highest % contribution) feature to predicting HON activity in our experimental setup (**Figure 7C**). Expanding this analysis across the frequency domain, an empirical mode decomposition of HON activity revealed that pupil size was maximally encoded at a lower frequency than body movement (**Figure 7D-E**). In contrast, ocular movements were encoded at a higher frequency than body movement. Overall, these results suggest that HON encoding of movement is distinct from their encoding of pupil-linked arousal.

## Movement tracking in HON versus other genetically-defined subcortical neural populations

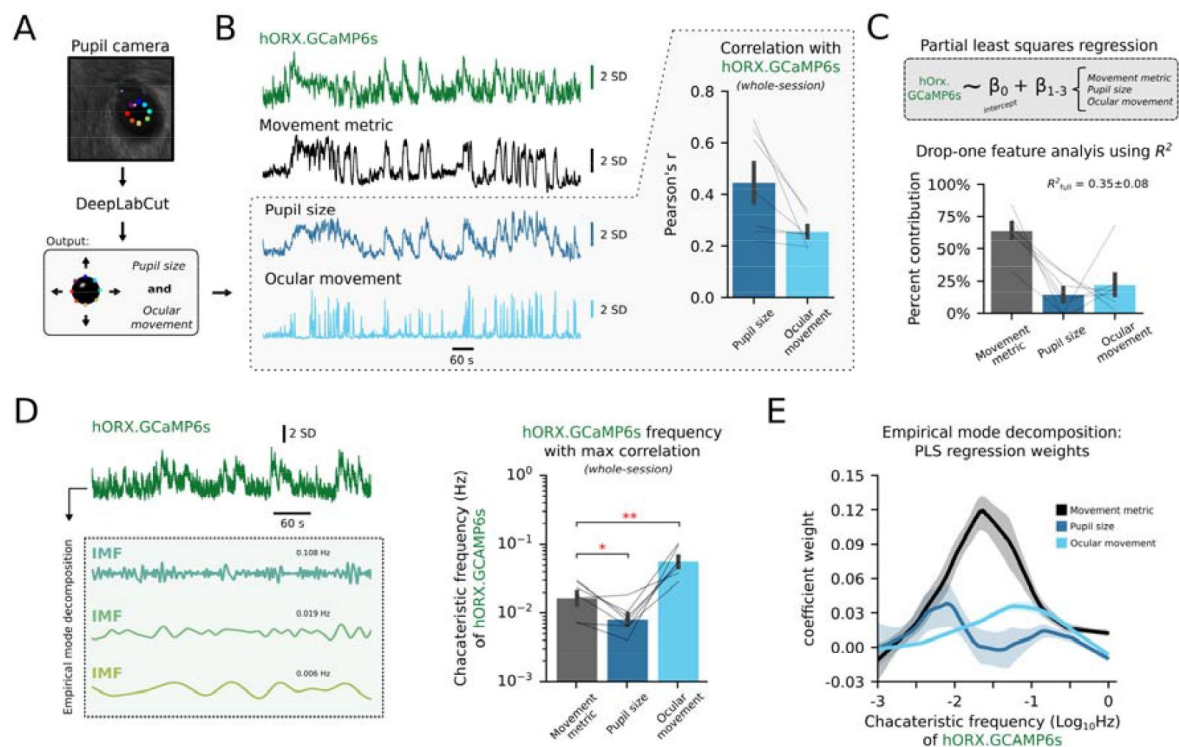
Finally, we sought to contrast the movement encoding in the HON population against other genetically defined subcortical neural clusters. We selected dopaminergic neurons of the SNc, glutamatergic neurons of the medial vestibular nucleus (MVe), and noradrenergic neurons of the LC, due to their links to HON and previously proposed roles in movement and/or arousal<sup>52–57</sup>. Additionally, we examined SF-1 expressing neurons in the ventral medial hypothalamus (VMH), which are thought to be sensitive to metabolic state<sup>58–62</sup>. We used different BL/6J-derived mouse cre-lines to express cre-dependent GCaMP6s activity indicators in each of these neural clusters, obtaining simultaneous fiber photometry neural activity and body movement recording in each mouse group, in the same way as for HONs (**Figure 1A**). We found that, like HONs ( $LHA^{ORX}$ ,  $r = 0.68 \pm 0.04$ ), the MVe glutamatergic neurons ( $MVe^{GLUT}$ ,  $r = 0.70 \pm 0.06$ ), correlated with movement across the entire recording (**Figure 8C**). The strength of the correlation was not significantly different across these two populations ( $p = 0.8366$ ). In contrast,  $LC^{NA}$  had only weak positive correlation ( $r = 0.23 \pm 0.07$ ), while  $SNc^{DA}$  had a weak negative correlation ( $r = -0.28 \pm 0.08$ ).  $VMH^{SF-1}$  neurons had only a slight positive trend ( $r = 0.20 \pm 0.11$ ). Cross correlation between GCaMP6s and the movement metric showed that HON activity on average preceded movement by a few samples ( $-0.34 \pm 0.12$  seconds), as did the  $LC^{NA}$  and  $SNc^{DA}$  ( $-0.72 \pm 0.15$  and  $-1.08 \pm 0.52$  seconds, respectively).  $MVe^{GLUT}$  activity was closely aligned to



**Figure 6.**

### HONs multiplex movement and blood glucose across frequency domains

**A**, Correlation of movement with HON photometry across three states of fasting ( $n = 15$  mice). Fed: ad-libitum access to chow. Re-fed: 18-hour fasted and then food returned 2 hours before experiment. Fasted: 18-hour fasted mice (rmANOVA:  $F_{2,28} = 1.848$ ,  $p = 0.176$ ). **B**, Scheme of experimental setup including photometry, glucose telemetry (DSI), video capture, and intragastric catheter infusions. **C**, Example HON GCaMP6s photometry trace (top), three derived IMFs (middle), and simultaneously recorded physiological variables of blood glucose (plotted as the first derivative) and movement magnitude (lower). A vertical dashed orange line represents an intragastric infusion of glucose. **D**, A linear model was fit to predict  $n = 94$  IMFs using the glucose derivative and movement. Fitted coefficient weights are plotted as a function of the IMF's characteristic frequency. **E**, Per-mouse ( $n = 6$ ) binned coefficient weights. For the higher frequency bin (left,  $10^{-1}$  to  $10^{-2}$  Hz), coefficient weights were tested against a population mean of zero (Bonferroni-corrected one-sample  $t$ -test: glucose derivative  $t_5 = 1.965$ ,  $p = 0.2131$ ; movement  $t_5 = 14.102$ ,  $p < 0.0001$ ). Coefficient weights of glucose derivative were compared to movement (paired  $t$ -test:  $t_5 = 8.331$ ,  $p = 0.0004$ ). For the lower frequency bin (right,  $< 10^{-3}$  Hz), coefficient weights were similarly tested against a population mean of zero (Bonferroni-corrected one-sample  $t$ -test: glucose derivative  $t_5 = -5.631$ ,  $p = 0.0049$ ; movement  $t_5 = 1.963$ ,  $p = 0.2139$ ). Coefficient weights of glucose derivative were compared to movement (paired  $t$ -test:  $t_5 = 5.385$ ,  $p = 0.0030$ ). Bar plots display mean  $\pm$  SEM. **F**, Alternate visualization of E with coefficient weights plotted against each other. **G**, Left: correlations of HON GCaMP6s signal and the movement metric across blood-glucose quartiles in an example mouse. Right: Regression predicting HON GCaMP6s activity using movement and blood-glucose percentile, performed per-mouse (interaction effect; one-sample  $t$ -test:  $t_5 = 0.6250$ ,  $p = 0.5594$ ). Bar plot displays mean  $\pm$  SEM. \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , \*\*\*\* $p < 0.0001$  and ns, not significant by two-tailed tests.



**Figure 7.**

### Disentangling contributions of arousal vs movement to HON activity

**A**, Scheme of pupillometry processing pipeline. **B**, Example aligned recording of HON GCaMP6s photometry (green), movement metric (black), pupil size (dark blue), and ocular movement (light blue). Right inset shows correlation of pupil size and ocular movement with HON photometry ( $n = 7$  mice). **C**, Partial least squares regression fit to predict HON GCaMP6s photometry from the movement metric, pupil size, and ocular movement. Percent contribution of each feature is quantified using drop-one feature analysis (see methods). **D**, Empirical mode decomposition of HON GCaMP6s IMFs to quantify the frequency of HON activity containing the maximum correlation with each feature. Movement metric maximally correlated with HON frequencies that were higher than that of pupil size, but lower than that of ocular movements (Bonferroni-corrected paired  $t$ -test: against pupil size  $t_6 = 3.4199$ ,  $* p = 0.0283$ ; against ocular movements  $t_6 = 4.6059$ ,  $** p = 0.0073$ ). **E**, Local regression displaying coefficient weights from a partial least squares regression fit to individual HON GCaMP6s IMFs, plotted as a function of the IMF's characteristic frequency.



movement ( $-0.15 \pm 0.09$ ), and VMH<sup>SF-1</sup> activity followed movement by several seconds ( $4.23 \pm 0.15$ ) using the  $1.5\times$  inter quartile range to filter for outliers (**Figure 8D**). We then repeated the clustering analysis described in **Figure 4**, aligning the photometry signals to initiation of movements assigned to “small” or “large” movements (**Figure 8E**). For large movements, HON activity encoded movement similarly to MVe<sup>GLUT</sup> neurons ( $p = 0.3915$ , **Figure 8G**). However, for small movements, HON activity encoded the movement metric significantly better than MVe<sup>GLUT</sup> neurons ( $p = 0.0099$ , **Figure 8G**). Together, these data suggest HONs are uniquely positioned to track body movement to a degree not observed in the other subcortical populations we examined.

## Discussion

We found that, across behaviors, HONs tracked the amount of body movement with a high degree of precision. This tracking occurred in a higher frequency bandwidth than their tracking of blood glucose, contributing to effective multiplexing of these two fundamental physiological variables. The HON movement tracking was not significantly different across hunger and glycemic states, suggesting orthogonal multiplexing of the movement and energy states. At two key projection targets of HONs, orexin/hypocretin peptide output was also related to movements, but in a projection-specific way, implying a heterogeneity in the movement-related HON outputs. Among the multiple neurochemically-defined subcortical neural clusters examined here, HONs were dominant in the precision of their movement tracking.

What would be the evolutionary advantages of coupling body movement to proportional changes in HONs? A central hypothesis for the meaning of HON activation is increased arousal, with evidence including electroencephalography<sup>26</sup>, pupil dilation<sup>50, 51</sup>, and cardiorespiratory adaptations<sup>63, 64–66</sup>. Though our projection-specific data imply that there may be subsets of HONs that are negatively related to movement (**Figure 5**), at the HON population level overall the coupling between neural activity and movement was positive ( $r=0.68$ , **Figure 1**). A simple interpretation of these data is that, via the HON movement tracking, the brain creates a “wake up” signal in proportion to movement, which could be helpful since dealing with challenges and opportunities created by movement may benefit from increased arousal. Importantly, HONs are also implicated in the mobilization of body glucose stores and shuttling glucose from liver to muscle<sup>67</sup>. Multiplexing their activation to both movement and glucose-sensing may allow movement process to be optimally matched to movement energy demands.

It is tempting to speculate on the breadth at which HON population activity encodes physiological derivatives (i.e. rate-of-change in physiological parameters) multiplexed across the frequency domain. This study highlights how HONs represent instantaneous movement magnitude, which can be rephrased as a change in body position over time. Similarly, we previously reported population HON activity predominantly tracks the derivative of blood glucose, rather than proportionally encode glycemic state<sup>23</sup>. Knowing the rate at which a physiological variable is changing allows for anticipatory responses, such as in Proportional-Integral-Derivative (PID) sensorimotor control loops, where the derivative-control component enables adaptive prediction of erroneous deviations from a set-point<sup>68, 69</sup>.

Do HONs “read” or “write” movements? From the perspective of sensorimotor control loops, the answer should be both. In theory, HONs could “read” movements either by sensing them, or by receiving efference copies and/or corollary discharges corresponding to motor commands<sup>70</sup>. Prior studies indicate that acute, targeted HON stimulation can evoke or modulate certain kinds of movement<sup>71–74</sup>. Orexin receptor antagonism – or optogenetic suppression or deletion of HONs – evoked either locomotor suppression or no effect on locomotor parameters<sup>75–77</sup>, consistent with our present result that the power-frequency spectrum of movements was not strongly dependent on presence of HONs themselves (**Figure 4**). Our finding that orexin level in the SNc – a site where orexin stimulates locomotion<sup>23</sup> – is higher before movements rather than





during movement (**Fig. 5**) also suggest that at this site, orexin may be more involved in movement initiation rather than maintenance. Interestingly, in the LC, the opposite orexin profile is seen (**Fig. 5**); given the role of LC in arousal, this may mean that orexin in the LC provides increased arousal during ongoing movement. Overall, in terms of movement control, HONs may be sufficient to “write” movements, but they are not necessary. Furthermore, it is tempting to speculate on the implications of HON loss-of-function viewed from the lens of a sensorimotor control loop. Without derivative-control, a system is still able to measure and perturb a variable, however its ability to dampen unwanted oscillations from a set-point is strongly impaired. From this perspective we can speculate that narcolepsy-cataplexy, caused by HON loss-of-function, is perhaps explained by oscillations into unwanted sleep-states and motor programs due to impaired control loops for wakefulness and movement.

In addition to these fundamental implications, there is an applied implication of our findings for basic neuroscience research. Ever since the importance of HONs for brain function and brain-body coordination was established<sup>24, 25, 31, 78</sup>, a demand appeared for tracking their function across research and clinical settings, and many useful (but invasive) techniques for this were developed<sup>40, 45, 47, 48, 79</sup>. The high correlation between body movement and HON activity demonstrated here suggests that video-tracked pixel difference could serve as a simple non-invasive biometric for real-time estimation of HON population state, at least as validated here in head-fixed behaving mice, a widely used model in neuroscience research<sup>80, 81</sup>, where assessment of arousal-linked signals is in continuing demand<sup>15, 82–84, 85, 86</sup>.

Our results open multiple directions for further study. Upstream of HONs, it would be important to solve the challenging puzzle of how movement information is conveyed to these neurons. This challenge may likely require an extensive functional screen, because of the wide-spread monosynaptic inputs to HONs, with many of these coming from areas where some movement-associated activity has been reported<sup>87</sup>. Within the HON population activity studied here, individual cells and their possibly heterogeneous responses to movement also remain to be defined, since the present results likely represent only a dominant subpopulation of HONs. Downstream of HONs, the findings in **Figure 5** could be extended to probe the role in movement communications of the multiple transmitters released by HONs<sup>88–90</sup> to the genetically-distinct, brain-wide targets of their terminal projections<sup>29</sup>. In future work, it will also be important to test whether movement tracking by HONs is used for optimal adjustments of the many cognitive and autonomic processes linked to HON outputs.

In summary, our findings define a key aspect of brain activity that is rapidly and precisely synchronized to even small alterations in body movement. The brain needs to sense movement for predicting future states and energy needs. Our results show that movement is encoded in parallel within glycaemic state within different frequency domains of HON activity, presumably allowing these cells to communicate both energy supply and a key aspect of energy demand in the body. Deeper insights of how defined neurons and connection turn body movement into adaptive responses will enable insights into the impact of body movement on brain function in health and disease.

## Materials and Methods

### Experimental Subjects

All animal experiments followed the Swiss Federal Food Safety and Veterinary Office Welfare Ordinance (TSchV 455.1, approved by the Zürich Cantonal Veterinary Office). Adult C57BL/6J-derived lines were studied. We used previously validated cre-dependent transgenic mice: Vglut2-ires-Cre (Slc17a6<sup>tm2(cre)Lowl</sup>/J, JAX:016963), SF1-Cre (Tg(Nr5a1-cre)7Lowl/J, JAX:012462), DAT-Cre (Slc6a3<sup>tm1(cre)Xz</sup>/J, JAX:020080), or DBH-iCre (C57BL/6-Tg(Dbh-icre)1Gsc, MGI:4355551,<sup>91</sup>). For

HON ablation experiments we used the previously validated orexin-DTR mice, wherein all HONs were ablated via injection with diphtheria toxin >2 weeks before experiments began [40](#), [41](#). All mice were male except for a small number of Vglut2-ires-Cre and DBH-iCre mice, which were similar to males and therefore pooled. Thus, whether all conclusions apply to female mice remains to be determined. Animals were housed in a 12:12 hour light-dark cycle and had ad libitum access to water and food (3430 Maintenance Standard diet, Kliba-Nafag) unless stated otherwise. All experiments were performed in the dark phase. Studies were repeated in at least two independent cohorts and, when relevant, used a semi-randomized crossover design for fasting state.

## Surgeries and Viral Vectors

To record photometry across a range of brain regions, we stereotactically injected viral vectors into adult mice as follows. In all cases, mice were anesthetized with 2-5% isoflurane following operative analgesia of buprenorphine and site-specific lidocaine. Viruses were stereotactically injected at 1-2 nL/s (NanoInject III injector) either unilaterally, or bilaterally, at the following coordinates relative to bregma: anteroposterior (AP), mediolateral (ML), and dorsoventral (DV).

The previously validated promotor-driven AAV1-hORX-GCaMP6s.hGH <sup>40</sup> ( $2.0 \times 10^{13}$  GC/mL, Vigene Biosciences) was used to record from HONs in the **LHA**: AP -1.35, ML  $\pm$  0.95, DV -5.70, -5.40, and, -5.10 mm. 70 nL per site. For cre-dependent mouse lines, we injected AAV9.CAG.Flex.GCaMP6s.WPRE.SV40, a gift from Douglas Kim & GENIE Project (Addgene plasmid # 100842;  $1.9 \times 10^{13}$  GC/mL, 1:5 dilution in sterile PBS) in the **MVe**: AP -6.05, ML  $\pm$  0.95, DV -4.40 mm. 200 nL per site. In the **VMH**: AP -1.35, ML  $\pm$  0.45, DV -5.75, -5.50, and -5.25 mm. 70 nL per site. In the **SNC**: AP -3.20, ML  $\pm$  1.40, DV -4.20. 200 nL per site. In the **LC**: AP -5.35, ML  $\pm$  0.90, DV -3.70 and -3.50 mm. 100 nL per site. The orexin-sensor, AAVDJ.hSynapsin1.OxLight1 ( $7 \times 10^{12}$  GC/mL, UZH Viral Vector Facility, 1:3 dilution in sterile PBS) was injected into both the SNC or LC at the above coordinates. 150 nL per site. Optic fiber cannula of 200  $\mu$ m diameter, 0.39 numerical aperture (NA) with a 1.25 mm ceramic ferrule (Thorlabs) were implanted 0.1 mm above the most dorsal injection site. A custom-made aluminum head-plate was secured to the skull using dental cement (C&B Metabond). Mice were given postoperative analgesia and allowed to recover for at least two weeks before the first experiment.

Expression was confirmed experimentally by observing dynamics at 465 nm excitation. A small number of mice without any observable dynamics at 465 nm, or with strong artifacts in the 405 nm isosbestic, were excluded from analysis. Expression was additionally confirmed terminally, via perfusion with 4% PFA in sterile PBS, histological slicing, and then imaging with a fluorescence microscope (Nikon Eclipse Ti2). For histological confirmation of the orexin-DTR mice, HONs were labeled with goat anti-orexin A (1:250 dilution, sc-8070 Santa Cruz Biotechnology) followed by and donkey anti-goat Alexa-546 (1:500 dilution, A11056 Invitrogen) and a Hoechst stain.

For intragastric infusion surgeries as in **Figure 6**, a small number of experiments from Viskaitis et al. [23](#) were re-analyzed if video was also available, the detailed methods for these glucose experiments are given in this source publication [23](#).

## Fiber Photometry

For fiber photometry, we used a custom-build camera-based photometry system. GCaMP6s (or OxLight1) activity was recorded at 10-20 Hz via alternating excitation between 405 nm and 465 nm LEDs (Doric; average power 20  $\mu$ W at fiber tip) in multiple animals simultaneously. The 405 nm excitation was used as an isosbestic control for movement artifacts. Most recordings were either 20 or 40 minutes long, except in **Figure 6** when recording lasted for >1 hour. The first minute of each photometry trace was removed due to bleaching artifacts. Further 465 nm fluorescence bleaching was accounted for by subtracting a triple exponential curve fit to the trace. Finally, each trace was z-score normalized to its standard deviation and mean. For visual clarity only,

photometry traces in [Figure 1D](#), [Figure 5A](#), [Figure 7B](#), and [Figure 8A](#) were smoothed using a 1-second moving average filter; all statistical comparisons and derived metrics were calculated using raw data.

## Movement Metrics

All experiments were performed in a closed container illuminated via infrared LEDs to achieve constant luminance during the recording. The movement metric was generated from videos captured using a commercial infrared camera (Blackfly S USB3, Teledyne FLIR) at the same sampling rate as photometry (10-20 Hz). To generate the metric, per-pixel differences were calculated across consecutive frames with 8-bit resolution. Then, the absolute value of the difference across every pixel was summed for each consecutive frame to generate one value for each sample, excluding the first frame. To compare this movement metric to concurrently measured photometry, the resulting trace was convolved with a “GCaMP6s kernel” which consisted of a 60 second exponential decay kernel with a decay rate equivalent to the reported GCaMP6s half-life (1.796 s<sup>38</sup>). Then, the first minute of convolution was removed to match the photometry trace and the resulting trace transformed to a z-score using its standard deviation and mean. To calculate the correlation between the movement metric and photometry, we aligned all samples from both outputs across the entire recording period, then calculated a Pearson’s *r* (scipy.stats.pearsonr, SciPy v1.14.0). Pearson’s *r* results were averaged per-mouse across repeated recording sessions. When relevant, locomotion-specific motion was recorded using an optical encoder (Honeywell, 128 ppr 300 rpm Axial) and digital acquisition box (BNC-2110, National Instruments). Locomotion was filtered using a 5-sample (250 ms) moving average filter for visualization only.

For bout-related analyses, we detected movement-bouts by first smoothing the z-scored movement-metric with a 20-sample (1 second) moving average filter and then aligning to sample indices where movement was < 0 SD for at least 4 seconds before exceeding the threshold. For subsequent clustering, a K-Means algorithm (scikit-learn) in the default configuration was implemented to identify *k* = 2 clusters of movement from the identified bouts. Photometry signals were aligned to bout indices then baselined by calculating a z-score of 1-3 seconds before bout initiation.

## Pupillometry

Pupil dynamics were recorded using a separate infrared camera (Blackfly S USB3, Teledyne FLIR) at a frame rate synchronized to photometry and the body-movement camera at 20 Hz. To calculate pupil metrics, we used a standard DeepLabCut<sup>92,93</sup> 3.0 pipeline and manually labeled 405 video frames to identify eight points surrounding the edge of the pupil. A model was fit following 5-percent holdout validation using default recommended DeepLabCut parameters for a pretrained ResNet50<sup>94</sup>. Briefly, the model was trained over 200 epochs from a starting learning rate of  $1 \times 10^{-4}$  using the Adam optimizer. The learning rate decreased by a factor of ten at epoch 160 and again at 190, following the default scheduler. After the model fit, we extracted pupil coordinates from all videos and manually inspected for accuracy. Using python, we fit a circle (scipy.optimize.minimize, method=“Nelder-Mead”) to all points which DeepLabCut assigned a confidence above 0.8. During blinking, or when DeepLabCut did not return a confidence of 0.8 for at least 3 points, pupil size was linearly interpolated. Pupil size was defined as the area of the fitted circle. Ocular movements were defined as displacements of the fitted circle’s center. Pupil size and ocular movements were then z-scored to the mean and standard deviation of the entire trace.



## Concurrent monitoring of glucose and intragastric infusions

The HD-XG telemetry system (DSI) as described in Viskaitis et al. <sup>23</sup> was used to measure and preprocess blood glucose concentrations as per manufacturer instructions. For intragastric (IG) glucose infusions, 900  $\mu$ L 0.24g/mL glucose dissolved in sterile saline was infused at a rate of 90  $\mu$ L/minute for 10 minutes. Saline controls are shown in the source publication and did not perturb HONs<sup>23</sup>. Missing samples were linearly interpolated, and the resulting glucose trace was resampled (`scipy.signal.resample`) to align to the photometry recording Hz. To generate the derivative traces, the raw glucose trace was smoothed using a 10 minute moving average filter then resampled to align to the photometry recording Hz before the first derivative was calculated.

## Empirical mode decomposition

The logic of empirical mode decomposition analysis was largely based on <sup>95</sup>, and was performed using the default settings of the python EMD toolbox version 0.6.2 <sup>96</sup>. Briefly, traces were first normalized (z-score) before IMFs were calculated using the sift algorithm <sup>39</sup>. The instantaneous phase, amplitude, and frequency across each IMF were calculated using the Normalized Hilbert Transform Method. The characteristic frequency was determined by taking the mean of each IMF's frequency data weighted per-sample by the IMF's amplitude data. Extremely low-frequency ( $< 10^{-3}$  Hz) IMFs were excluded as they did not complete enough cycles to allow sufficient sampling for the following statistics. Power was taken as the means of the squared amplitude data and expressed as a percentage of the total power from each IMF. The preferred phase (direction) and coupling strength (length) for each IMF's phase data was then calculated from the 1<sup>st</sup> trigonometric circular moment with weights corresponding to positive ("active") or negative ("quiescent") signed samples of the movement metric. Note that in the "active" case, for example, all negative values of the movement metric would be assigned weights of zero.

## Behavioral classification

To classify behaviors, we used the video captures and manually labeled five behavioral classes: chewing, grooming, resting, running, and sniffing, from greyscale frames using the previous, current, and subsequent frames to form an RGB input image. 4'221 manually labeled frames were split into a "train" and "test" datasets, stratified across behavioral classes, following ten-percent holdout validation. When generating the datasets, we augmented the dataset using random horizontal and vertical flips of the input frames, to ensure robustness of the behavioral classification. A convolutional neural network was constructed using PyTorch 2.2.0: we used the standard ResNet18 with pretrained weights on the ImageNet dataset as previously described <sup>94</sup>, but with a final linear layer corresponding to the five output classes. To train the model, we used an Adam optimizer with an initial learning rate of  $1 \times 10^{-4}$ , and a cross entropy loss function. A scheduler (`torch.optim.lr_scheduler.StepLR`) decreased the learning rate by a factor of 10 every 10 epochs. The model was fit on 41 total epochs using a Tesla T4 GPU (NVIDIA) using a batch size of 128. The model was independently fit several times and consistently achieved over 98.6% accuracy on the "test" dataset, with a cross entropy loss of  $< 0.054$ . The trained network was then used to classify photometry-synchronized videos. The model output was passed through a softmax function, and thereby consisted of  $n$  video-frames  $\times$  5 behavioral likelihoods. We smoothed the model-predicted likelihoods for each behavior using a 4-second moving average filter and took the max-likelihood behavior at each frame as the assigned classification. Finally, the videos and predictions were manually inspected for accuracy.

## Data analysis and statistics

Raw data were processed with custom scripts in Python 3.9, using Seaborn and Matplotlib libraries for plot-generation. Statistical analysis was similarly performed in Python using Pingouin v0.5.4, SciPy v1.14.0, and scikit-learn 1.2.1 libraries. In **Figure 2** and **3**, local regressions were performed using statsmodels v0.14.1 with a 0.3-0.5 fraction of the data. Confidence intervals were

generated using 1000 bootstrapped samples. In the case of strictly positive variables (% power), we used a logarithm link function and transformed the result back into its original units for visualization. Circular regressions were similarly performed using scikit-learn and a multi-output support vector regression whereby the phase ( $\theta$ ) in radians was evaluated over 7 evenly spaced samples on the log-scale frequency by converting the phase into a multidimensional vector  $y$ :

$$y = [\sin(\theta), \cos(\theta)]$$

Linear mixed effects models in **Figure 2E** were used to predict z-scored GCaMP6s photometry using the z-scored movement metric or behavioral classification. Models were defined as follows:

$$\text{Model 1, } GCaMP6s_{ij} = \beta_0 + \beta_1 \times C(\text{Behavior})_{ij} + b_{0i} + \epsilon_{ij}$$

$$\text{Model 2, } GCaMP6s_{ij} = \beta_0 + \beta_1 \times \text{Movement}_{ij} + b_{1i} \times \text{Movement}_{ij} + \epsilon_{ij}$$

$$\text{Model 2, } GCaMP6s_{ij} = \beta_0 + \beta_1 \times \text{Movement}_{ij} + b_{1i} \times \text{Movement}_{ij} + \beta_2 \times C(\text{Behavior})_{ij} + b_{0i} + \epsilon_{ij}$$

Where  $GCaMP6s_{ij}$  is the photometry sample,  $C(\text{Behavior})_{ij}$  is the behavioral classification, and  $\text{Movement}_{ij}$  is the movement metric for the  $j$ -th observation of the  $i$ -th mouse.  $\beta_0$  is the intercept, and  $\beta_1$  and  $\beta_2$  are the fixed effects.  $b_{0i}$  is a random intercept of the  $i$ -th mouse.  $b_{1i}$  is the random slope for movement of the  $i$ -th mouse.  $\epsilon_{ij}$  is the residual error. We assumed the residual error was normally distributed.

Linear regression models for **Figure 6** were calculated using the statsmodels library following the formula:

$$IMF_n = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$$

Where  $IMF_n$  is the  $n^{\text{th}}$  IMF of the GCaMP6s signal,  $\beta_0$  is the intercept term,  $\beta_1$  and  $\beta_2$  are the coefficients for  $X_1$ , blood-glucose derivative, and  $X_2$ ; movement metric, respectively, and  $\epsilon$  is the error term.

Partial least squares regressions in **Figure 7** were calculated using scikit-learn with  $n = 2$  components. To perform drop-one feature analysis<sup>97</sup>, the movement metric, pupil size, and ocular movements were used as predictors to generate a “full model.” The generated  $R^2$ -score (calculated as the average of a 5-fold cross validation) is denoted as  $R^2_{\text{full}}$ . The predictors were then removed from the model one at a time, and a “partial”  $R^2_{\text{partial}}$  was generated. The contribution of each predictor  $k$  was therefore defined as:

$$\% \text{ Contribution}_k = 100 \times \left( 1 - \frac{R^2_{\text{partial},k}}{R^2_{\text{full}}} \right) / \sum_{j=1}^3 \left( 1 - \frac{R^2_{\text{partial},j}}{R^2_{\text{full}}} \right)$$

Statistical tests, sample sizes, and their results are indicated in the figures, legends, and/or descriptions in the text. Where significance is presented, p-values are as follows: \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , and \*\*\*\* $p < 0.0001$ , n.s.  $p > 0.05$ . We made no assumptions regarding the directionality of our effects; all statistical tests were two-tailed. When relevant, multiple comparisons adjustments were made using the Bonferroni correction as indicated in figure legends / or text. Data are presented as means and standard error of the mean (SEM) unless stated otherwise. Boxplots always represent the maximum, minimum, interquartile range, and median of the data.

## Data availability

Source data has been deposited in a publicly available database (<https://osf.io/6d45m/>).

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### Reviewer #1 (Public review):

Summary:

This manuscript by Tesmer and colleagues uses fiber photometry recordings, sophisticated analysis of movement, and deep learning algorithms to provide compelling evidence that activity in hypothalamic hypocretin/orexin neurons (HONs) correlates with net body movement over multiple behaviors. By examining projection targets, the authors show that hypocretin/orexin release differs in projection targets to the locus coeruleus and substantia nigra, pars compacta. Ablation of HONs does not cause differences in the power spectra of movements. Movement tracking ability of HONs is independent of HON activity that correlates with blood glucose levels. Finally, the authors show that body movement is not encoded to the same extent in other neural populations.

Strengths:

The major strengths of the study are the combination of fiber photometry recordings, analysis of movement in head-fixed mice, and sophisticated classification of movement using deep learning algorithms. The experiments seem to be well performed, and the data are well presented, visually. The data support the main conclusions of the manuscript.

Weaknesses:

To some degree, it is already known that hypocretin/orexin neurons correlate with movement and arousal, although this manuscript studies this correlation with unprecedented sophistication and scale.

Taken together, this study is likely to be impactful to the field and our understanding of HONs across behavioral states.

<https://doi.org/10.7554/eLife.103738.2.sa3>

### Reviewer #4 (Public review):

Summary:

Using head-fixed approach, the authors show a rapid impact of movement on the activity level of hypothalamic orexin/hypocretin neurons.

Strengths:

The head-fixed approach is great to isolate specific movements and their impact on neuronal activity.

Weaknesses:

Many of the weaknesses that were noted in the previous round of review have been addressed.

<https://doi.org/10.7554/eLife.103738.2.sa2>

**Reviewer #5 (Public review):**

**Summary:**

Hypothalamic hypocretin/orexin neurons are well-known to be involved in arousal, muscle tone and energy metabolism. Using a combination of fiber photometry, video-based movement assessments, and deep learning algorithms, the authors provide compelling evidence that the activity of these neurons correlates with net body movement over multiple behaviors and is independent of nutritional state. The authors also demonstrate that hypocretin/orexin release differs between two downstream projection sites, the locus coeruleus and substantia nigra, and are able to distinguish the activity in these sites that is due to inputs from these hypothalamic neurons vs. from other subcortical populations. The authors also convincingly show that the correlation between body movement and hypocretin/orexin neuron activity is much stronger compared to other subcortical regions. However, hypocretin/orexin neuron ablation does not affect the power spectra of movements, an observation that appears at odds with their overall conclusions.

**Strengths:**

The multidisciplinary approach using multiple state-of-the-art tools is supported by a rigorous experimental design and strong statistical analyses. The authors have been highly responsive to previous critiques. Concerns of another reviewer regarding the confound between arousal and movement have been addressed by new pupillometry data as a measure of arousal and multivariate analyses to distinguish between the contributions of arousal vs. movement to hypocretin/orexin neuron activity. The new data in Figure 2H added in response to a suggestion by Reviewer 3 particularly strengthens the paper.

**Weaknesses:**

Reviewer 2 mentioned that previous studies using orexin antagonists in rodents have largely found inconsistent effect of antagonizing orexin signaling on simple motor activity and points out that these studies are not referenced here. The authors respond that "orexin antagonism - or optogenetic silencing of HONs - evokes either reduced locomotion, or no effect on locomotor movements" and add references to paragraph 4 of the Discussion. Aside from the fact that 2 of the 3 references added are from the senior author, none address the fact that orexin antagonists induce sleep and that optogenetic silencing of these cells creates a condition where sleep can ensue with short latency - results that certainly affect body movement/locomotor activity.

<https://doi.org/10.7554/eLife.103738.2.sa1>

**Author response:**

The following is the authors' response to the original reviews

**Reviewer #1 (Public review):**

**Summary:**

*This manuscript by Tesmer and colleagues uses fiber photometry recordings, sophisticated analysis of movement, and deep learning algorithms to provide compelling evidence that activity in hypothalamic hypocretin/orexin neurons (HONs) correlates with net body movement over multiple behaviors. By examining projection targets, the authors show that hypocretin/orexin release differs in projection targets to the locus coeruleus and substantia nigra, pars compacta. Ablation of HONs does not cause differences in the power spectra of movements. The movement-tracking ability of HONs is independent of HON activity that correlates with blood glucose levels. Finally, the authors show that body movement is not encoded to the same extent in other neural populations.*

**Strengths:**

*The major strengths of the study are the combination of fiber photometry recordings, analysis of movement in head-fixed mice, and sophisticated classification of movement using deep learning algorithms. The experiments seem to be well performed, and the data are well presented, visually. The data support the main conclusions of the manuscript.*

We thank the reviewer for their supportive feedback.

**Weaknesses:**

*The weaknesses are minor, mostly consisting of writing and data visualization throughout the manuscript. To some degree, it is already known that hypocretin/orexin neurons correlate with movement and arousal, although this manuscript studies this correlation with unprecedented sophistication and scale. It is also unfortunate that most of the experiments throughout the study were only performed in male mice. Taken together, this study is likely to be impactful to the field and our understanding of HONs across behavioral states.*

We agree that disentangling movement from arousal is an important aspect, and in the revised manuscript, we now include new data and analyses towards this (pupillometry to directly assess arousal, and multivariate analysis to assess contributions of arousal vs movement to HON activity). In addition, we now implement many of the reviewer's recommendations regarding writing, data presentation, and visual clarity (see our replies in the "recommendations for authors" section).

**Reviewer #1 (Recommendations for the authors):**

*Some recommendations for the authors:*

*(1) The first sentence of the Introduction states: "Neural activity related to body movement recently received much attention." I would rephrase or clarify this statement, as neuroscientists have been studying neural activity related to body movement for decades.*

The reviewer is correct. Our intention was to highlight the resurgence of movement-related neurosciences enabled by modern techniques such as deep learning applied to video data (e.g. DeepLabCut, etc). The passage has been updated for clarity.

*(2) The Introduction also states that HONs orchestrate "consciousness and arousal." I would delete the word "consciousness," as consciousness represents a lofty, global concept that is challenging to define and quantify in humans, let alone mice.*

We used the word consciousness to be consistent with current literature on the function of the mouse hypothalamus (e.g. Nat Neurosci 2016 Feb;19(2):290-8). But we agree it is not necessary here, and so we followed the advice to delete it.

*(3) The authors state that HON dynamics were recorded while mice were head-fixed while on a running wheel. For clarity, it would be helpful to visualize this head-fixation in Figures 1A and 5B. It would also be helpful to clarify how certain behaviors (e.g. grooming, chewing) were performed and recorded while the mouse was head-fixed.*

In the revised manuscript, updated graphics with a head-fixed mouse have now been added to relevant figures. Representative RGB frames (colors representing sequential frames) of each behaviour have been added to Figure 2A.

*(4) In the legend for Figure 1A, the reference to Gonzalez et al. 2016 seems out of place (at least the reader should be informed why the text is referring to this previous study). Additionally, because the references are ordered by number instead of alphabetically, it would be more helpful to refer to a numbered reference rather than a name.*

Gonzalez et al. 2016 references the source of the AAV construct used in this figure. This has been moved to the methods. Following eLife formatting guidelines, references will be alphabetized upon publication.

*(5) In Figure 3F, it would be helpful to show visual validation that the HON-DTR method indeed ablates all HONs. This is depicted conceptually, but representative figures would be much more convincing.*

A representative histological slice is now included for both wild type (WT) and HON-DTR mice in the new Figure 4B.

#### **Reviewer #2 (Public review):**

##### *Summary:*

*Despite several methodological strengths, the major and highly significant drawback is the confound of arousal with movement. This confound is not resolved, so the results could be explained by previously established relationships between orexin and arousal/wakefulness.*

This an excellent point, and we agree. To address this directly in the revised manuscript, we now include new data and analyses towards this (pupillometry to directly assess arousal, and multivariate analysis to assess contributions of arousal vs movement to HON activity).

##### *Strengths:*

*The authors show that orexin neuron activity is associated with body movement and that this information is conveyed irrespective of the fasted state. They also report differences in different orexin target brain regions for orexin release during movement. This paper contains an impressive array of cutting-edge techniques to examine a very important brain system, the orexin-hypocretin system. The authors offer an original perspective on the function of this system. The authors showed that orexin neuron activity scales to some degree with the magnitude of body movement change; this is unaffected by a fasted state and seems to be somewhat unique to orexin neurons.*

*The investigation of other genetically defined subcortical neuron populations to determine the specificity of findings is also a strength, as is the ability to quantify*



*movement and use deep learning to classify specific behaviors adds sophistication to analysis. The authors also show heterogeneity in orexin projections to specific target nuclei, which is interesting.*

*The authors "speculate that narcolepsy-cataplexy, caused by HON loss-of-function, is perhaps explained by oscillations into unwanted sleep-states and motor programs due to impaired control loops for wakefulness and movement". This is quite an interesting aspect of their work and deserving of further study.*

We thank the reviewer for their supportive feedback.

*Weaknesses:*

*Despite the strengths, there are several major and minor weaknesses that detract significantly from the study.*

*My main concern with this work is the confound of arousal with movement so that correlations with one might reflect a relationship instead with the other. The orexin system is well known to play an important role in arousal, with elevated activity of orexin neurons reported for waking and high arousal. Orexin signaling has also been strongly associated with motivation, which also is associated with arousal and movement. The authors offer no compelling evidence that the relationships they describe between different movements and orexin signaling do not simply reflect the known relationship between arousal and motivation.*

*The authors could address this concern by including classical arousal measurements, eg, cortical EEG recorded simultaneously with movements. Often, EEG arousal occurs independently of movement, so this could provide one approach to disentangling this confound. The idea that orexin signaling plays a role in arousal rather than movement is supported by their finding that orexin lesions using the orexin-DTR mouse model did not impact movements. In contrast, prior lesion and pharmacologic studies have found that decreased orexin signaling significantly decreases arousal and waking.*

*Another way they could test their idea would be to paralyze and respiration animals so that orexin activity could be recorded without movement. Alternatively, animals could be trained to remain motionless to receive a reward. Thus, there are several ways to test the overall hypothesis of this work that have not been examined here.*

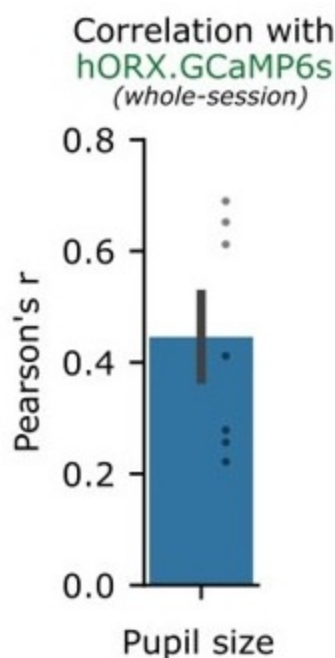
*The authors propose that "a simple interpretation of their results is that, via HON movement tracking, the brain creates a "wake up" signal in proportion to movement". This seems to argue for the role of the orexin system in arousal and motivation rather than in movement per se.*

Thank you. We agree that disentangling between arousal and movement is indeed critical. A classic approach is a multivariate analysis, wherein multiple simultaneously recorded "predictors" of HON activity – such as arousal and movement – can be directly compared. While EEG arousal is an option, another well-accepted metric for arousal is pupil diameter. Using  $n = 7$  mice, we now simultaneously record HON activity, movement, running speed, pupil size fluctuations, and ocular movements:

We then fit a partial least squares multivariate regression (a regression type more robust to collinearity) using the movement metric, pupil size, and ocular movements as predictors of orexin neuron activity. Consistent with previous publications, we found that pupil size alone has a positive correlation with hORX.GCaMP6s ( $\sim 0.45$ ). However, using a drop-one feature analysis in multivariate regression, we found that movement had the highest % contribution

to statistically explaining orexin neuron activity. Here are the new results (which we now added as Fig. 7A-B).

**Author response image 1.**



Furthermore, we also expanded this analysis to incorporate the different frequencies found in HON dynamics, using empirical mode decomposition. We found that pupil size had a maximum correlation at lower HON frequencies than the movement metric, while ocular movements were maximally correlated in higher frequencies (now added as Fig. 7D,E).

Overall, this analysis suggests that – while HONs encode both movement and arousal – arousal and movement do not always co-fluctuate at the same timescales, and their impacts on HONs can be disentangled in a number of ways. We now mention this in revised text on page 5.

*There are several studies that have examined the effect of orexin antagonist treatment in rodents on locomotor and other motor activities. These studies have largely found no consistent effect of antagonizing orexin signaling, especially at the OxR1 receptor, on simple motor activity. These studies are not referenced here but should be taken into account in the authors' conclusions.*

We agree. Prior studies found that orexin antagonism – or optogenetic silencing of HONs – evokes either reduced locomotion, or no effect on locomotor movements. We now added text and references to paragraph 4 of Discussion, summarising this.

*Figure 3, panel F: I understand HON-DTR is a validated model but a picture of HONs ablation is necessary, including pictures of HONs outputs ablation within the SNc and LC.*

A representative histological slice is now included for both wild type (WT) and HON-DTR mice in the new Figure 4B. Because HONs are only found in the hypothalamus, somatic deletion of HONs in this region will result in axonal degradation in output regions.

*The discussion lacks a more extensive paragraph on the distinct signal and role of Ox>SNc and Ox-LC projections.*

We now added sentences discussing potential implications of this to Discussion (middle of paragraph 4).

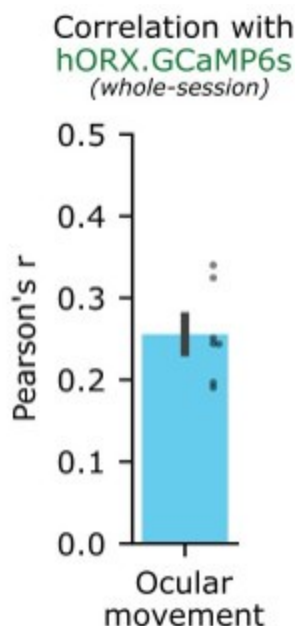
**Reviewer #2 (Recommendations for the authors):**

*Minor weaknesses*

*A very important movement in rodents is head orientation, especially given the limitation in ocular movement. However, this paper used a fixed head model which obviated this movement and did not attempt to analyze ocular movements.*

Analysing ocular movements is something we had not considered but is very easy to check using pupillometry. In  $n = 7$  mice, we recorded both orexin neurons, and ocular movements captured through an infrared camera under constant lighting. Ocular movements had a small positive correlation with orexin neuron photometry ( $r = \sim 0.26$ ). See response to the public review above.

**Author response image 2.**



*The "HON" abbreviation is not commonly used for orexin neurons, and I suggest replacing that with a more well-known abbreviation.*

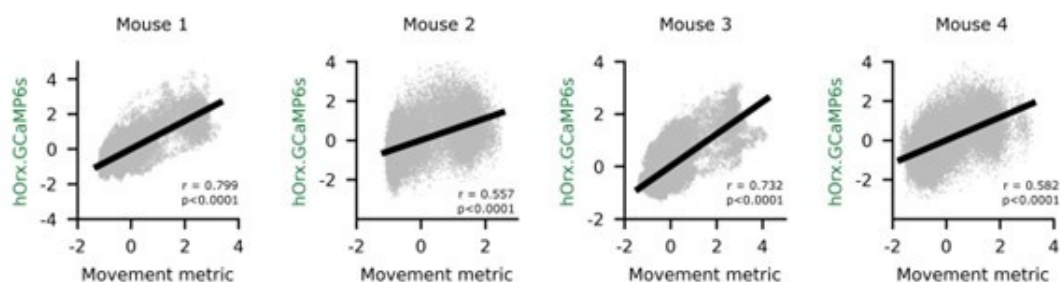
To the best of our knowledge, there is no universally agreed or best-known abbreviation for hypocretin/orexin neurons (we agree it would be nice if there was one!). "HONs" is a simple first letter abbreviation of hypocretin/orexin neurons, which acknowledges the two names for this peptide given by the original discoverers (de Lecea et al, and Sakurai et al, in 1998). Although this may not be the perfect abbreviation, we have kept it for now, also to be consistent with the large number (>10) of other published studies that recently used this abbreviation.

*The graphs showing Pearson's  $r$  values do not demonstrate a very strong correlation between neural activity and movement change; they also lack validation of genetic expression/ablation in some cases. The results would more strongly support the conclusions if statistically significant correlations could be demonstrated between activity and movement.*

We agree that a correlation of  $\sim 0.68$  is probably not worthy of a “very strong” classification. While there is no universal ruleset for categorizing the strength of a correlation, we have toned down our language throughout the manuscript.

Comment regarding statistical testing of correlations: we are cautious to stand behind correlation significance testing for large sample sizes ( $\sim 48'000$  photometry & video samples in a 40-minute session). In our case, correlations were always extremely significant  $p < 0.0001$ . The reason for this is that correlation  $p$ -values become “too big to fail” (see Lin et al. 2013) with inflated sample size. We therefore refrain from commenting on  $p$ -values and rather report between or within-subjects statistical tests, or tests against zero. See four example experiments below.

### Author response image 3.



Citation: Lin, M., Lucas, H. C., Jr & Shmueli, G. Research Commentary—Too Big to Fail: Large Samples and the  $p$ -Value Problem. Information Systems Research 24, 906–917 (2013).

*The rationale for looking at running speed, general movement, and specific types of nonlocomotor movements could be clarified and explained more thoroughly in the introduction. Why is it important to distinguish between locomotion (represented here with running) and all other movements? Presumably, this is because orexin is known to regulate arousal/locomotion. What evidence is there for orexin's role in other types of movements, which are being grouped together in Figure 1? This could be laid out in more detail in the Introduction. Relatedly, it is not very clear in the text whether the correlation between movement and orexin neuron activity includes movement related to running.*

The main focus of our paper is on movement in general (i.e. video pixel difference, described in Results and Methods). This movement metric includes everything captured by the video, it is agnostic to the type of movement or behaviour. To connect this to some of the specific innate movements/behaviours typically studied in mouse literature (running, grooming, sniffing, etc), we also performed plots in Figure 2. We attempted to explain this better in revised section 1 of Results.

*What exactly is being correlated in Figure 1C (and throughout the rest of the paper?) Is this the average signal correlated with the average movement change over the entire recording time? This could be more explicitly stated in methods/results. The correlations*

*themselves/p-values could be shown in addition to/instead of Pearson's  $r$  values. Are the correlations themselves significant? This would strengthen the claim that orexin activity is strongly coupled to the magnitude of body movement change. As another example, in Figure 2D, there are no statistics reported on the correlation between movement metric and average neural signal. In Figure 6G, orexin neuron activity is more strongly correlated with movement than MVe glut neurons, but are either of these correlations significant? The correlation between MVe glut activity and movement overall seems similar to that of orexin neurons, and may be worth noting more explicitly.*

Throughout the paper, we have recorded both neural activity (photometry) and movement at 20 Hz. This would generate, for example, 48'000 samples of photometry and movement from a 40-minute session. All the samples were used to calculate a pearson's  $r$  between variables. To clarify this, we now added the subtext "wholesession" to relevant figures, as well as a clarification in the methods.

Individual experiment correlations for orexin neurons and MVe glut neurons were always significant  $p < 0.0001$ , even after a Bonferroni multiple comparisons correction was applied to each population. See the "too big to fail" nature of correlation hypothesis testing above.

*It could be made clearer at the end of Figure 2 that orexin neuron activity is tracking the magnitude of movement change (shown in Figure 2D), not that it is encoding different types of movement.*

We intended for original Figure 2E to illustrate this concept, however this panel has caused a great deal of confusion to several readers and was perhaps ill conceived. We have replaced Figure 2E with a new panel more directly addressing the reviewer's statement. We can construct three models where orexin neuron activity is predicted from the behavioral classification (sometimes called "one-hot" encoding) and/or the movement metric.

Model 1 predicts orexin neuron activity using only a categorical predictor of behavioral state. Model 2 only uses the movement metric, and model 3 allows a different movement-metric correlation within each behavioral state. We can compare these models using AIC (Akaike Information Criterion) which is a point estimate. While the most complex model 3 was the best, model 2 was much closer to model 3 than model 1. Similarly, model 2 was much better than model 1. From this we conclude that the magnitude of movement change is a more powerful predictor than behavioral state ("type of movement"). This is now Figure 2E.

*It would be interesting to see the raw movement metric data as shown in Figures 1 and 2 in the DTR mice to show that ablating orexin neurons does not impair the movement profile seen in Figures 1 and 2.*

The requested visualization has been added to Figure 4B.

*Validation that orexin was selectively ablated in these mice would be ideal.*

Histology (see response to public review) was added to a new Figure 4B.

*Figure 4A - OxLight expression in SNc does not look very robust.*

Please note this is a membrane-targeted indicator, the staining this produces is thus much weaker than cytosolic indicators such as calcium indicator GCaMP.

*Figure 4 - It would be beneficial to see the same correlations that were done in Figures 1 and 2 to show OxLight activity vs. movement metric. Are they correlated?*



Individual traces had significant correlations with OxLight and movement, and the population averages revealed similar trends:

**Author response image 4.**

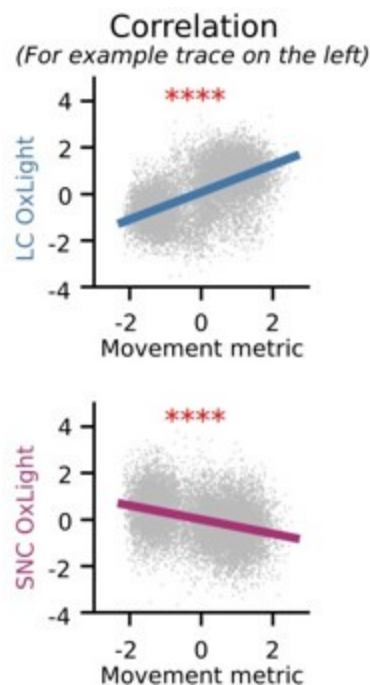


Figure 6B - Targeting of MVe neurons does not look very specific. The sample size for orexintargeted mice should be re-stated in the figure legend for clarity.

Legend has been updated to clarify n = 15 for orexin targeted mice.

Some citations didn't seem to match what was being referenced in the text. Similarly, in the legend for Figure 1C, the statistics do not match what is reported in the text. In Figure 1, the sample size is not noted in the text. When referring to running in Figure 1, is this referring to running speed? Perhaps the language could be more consistent.

These typos (due to a rounding error) in the legend and text have been corrected. Sample size has been added to the text, and we have changed Figure 1D to clarify we are referring to running speed. We moved some citations to improve clarity.

Methods - where were Cre mice obtained from?

Sources now better referenced in Methods (JAX or Parlato et al).

Figure 1, panel C: The authors compared Pearson's r-coefficient results for each animal and for each variable. However, it would be interesting to show the correlation curves for each variable. However, it would be interesting to show the correlation curves for each variable as well here. Also, there is mention of a strong correlation but it is unclear whether these correlations are significant.

See below for an example mouse.

Author response image 5.

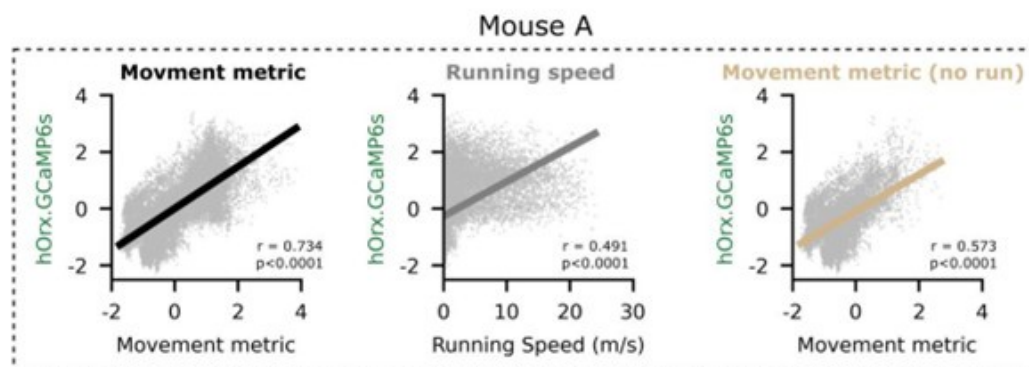


Figure 3, panel F: I understand HON-DTR is a validated model but a picture orexin ablation is necessary, including pictures of orexin fibers ablation within the SNc and LC.

See our reply to the public review above.

Figure 5, Panel A: Same comment as Figure 1, panel C.

We have similarly clarified the panel and legend.

Page 4: The authors mention "Within the 1st and 4th quartile of blood glucose, movement-HON correlations were not significantly different. Please add the figures.

The requested plot has been added to Figure 6, panel G.

**Reviewer #3 (Public review):**

*Summary*

The study presents an investigation into how hypothalamic orexin neurons (HONs) track body movement with high precision. Using techniques including fiber photometry, video-based movement metrics, and empirical mode decomposition (EMD), the authors demonstrate that HONs encode net body movement consistently across a range of behaviors and metabolic states. They test the ability of HONs to track body movement to that of other subcortical neural populations, from which they distinguish HONs activity from other subcortical neural populations.

*Strengths:*

The study characterizes HONs activity as key indicators of movement and arousal, and this method may have potential implications for understanding sleep disorders, energy regulation, and brain-body coordination. Overall, I think this is a very interesting story, with novel findings and implications about sensorimotor systems in animals. The manuscript is clearly written and the evidence presented is rigorous. The conclusions are well supported by experimental data with clear statistical analyses.

We thank the reviewer for their supportive feedback.

*Weaknesses/suggestions:*

*There are a couple of issues I think the authors could address to make the paper better and more complete:*

*(1) The study primarily focuses on steady-state behaviors. It would be interesting if the authors' current dataset allows analyses of HON dynamics during transitions between behavioral states (e.g., resting to running or grooming to sniffing). This could provide additional insights into how HONs adapt to rapid changes in body movement.*

This is a fantastic idea, and easy to check using our classification CNN. We identified the six most frequent behavioral transitions and plotted them in Figure 2H. HONs show rapid dynamics in activity aligned with behavioral changes.

These changes are very similar to the movement magnitude along these transitions, which is now also plotted in Figure 2G.

*(2) Given the established role of HONs in arousal and wakefulness, the study could further investigate how movement-related HON dynamics interact with arousal states. For example, does HON encoding of movement differ during sleep versus wakefulness?*

To further investigate how movement encoding interacts with arousal, we now include quantification and analysis of pupil-linked arousal (see new Figure 7). We agree it would be interesting to look at what happens during sleep, especially REM sleep when some HONs are thought to be active where there is no/little body movement, but this is beyond the scope of the present study.

*(3) Although HON ablation experiments suggest that HONs do not shape movement frequency profiles. It would be more compelling if the authors could investigate whether HONs contribute to specific types of movements (e.g., fine motor vs. gross motor movements) or modulate movement initiation thresholds.*

We performed this analysis using the k-means classifier for small/large movements. Consistent with previous results, we found no significant effect ( $p = 0.2767$ ) of genotype on the frequency of identified small (fine) or large (gross) movement clusters. This plot has been added to Figure 4E.

*(4) The heterogeneous movement-related orexin dynamics observed in the LC and SNc raise intriguing questions about the circuit-level mechanisms underlying these differences. Optogenetic or chemogenetic manipulation of these projections could validate the functional implications of these dynamics.*

We agree. We now discuss some implications of this in revised Discussion (paragraph 4). Please note that previous work already demonstrated that orexin action in the SNc can produce locomotion (referenced in the paragraph), though we agree that further work would be valuable.

### **Reviewer #3 (Recommendations for the authors):**

*Additional feedback:*

*(1) Figure 1C: the individual data points are hard to track or see. Consider using a larger marker face to help data visualization. Similar issues can be found in Figures 2C, 2E, 5E, 6C, 6F, and 6G.*

Thickness of the lines and scatterplots have been increased.

*(2) First Section of Results: the authors claim to use a deep-learning network to automatically classify video recordings into five distinct behaviors. However, several issues need to be addressed here:*

*a. In Results, the corresponding sentence lacks a reference to the Methods Section.*

Reference has been added to the text.

*b. In Methods, the description of the CNN model is quite limited, lacking many basic, necessary components including necessary references to published papers, the model training, characterization (only an overall accuracy is not enough), as well as dataset definition, preparation, augmentation (if any), etc.*

We have expanded the methods section regarding the CNN model.

*(3) First Section of Results: in the second paragraph, the authors claim that "Overall, these results reveal HON population activity precisely tracks a general degree of body movement across recorded behaviors." This is not accurate. To indicate that HONs activity tracks the general degree of body movement across behavior states, they need to further show that behavioral states with similar levels of movement metrics can be differentiated via HON activities. However, as they showed in Figure 2D, some behaviors with similar values of movement metric do not seem to be easily discerned by HON activity levels.*

We agree with you, and this is also what we originally intended to convey – now reworded for clarity.

*(4) Technical issue: Figures 3B, 3C, 3G, using local regression to plot the solid lines makes them touch negative values, which does not make sense for "power proportion" (this quantity is always non-negative).*

This is a good point. To fix this, we first log-transformed the power metric, then performed a local regression, and used the link function to transform the model predictions back to %-units for visualization. This has been noted in the methods.

*(5) Figure 3G: For a better comparison, consider combining the two plots into a single plot.*

The two plots have been merged as shown in Figure 4C.

*(6) Figure 5E: For a better data visualization, the current pair of plots can be consolidated into one single plot where the x-axis is Move and the y-axis is dGlu. In this way, it is easier to understand and the orthogonality as claimed in the manuscript can be more apparent.*

The requested plot has been added as Figure 6F.

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